

VIRGINIA BEACH BUOY 44014, USA

WMO: 996300

Lat: 36.606N

Lon: 74.840W

Elev: 0

StdP: 101.33

Time Zone: -5.00 (W05)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.8	0.0	-9.3	1.7	-0.8	-7.7	2.0	1.1	18.0	3.7	16.5	4.4	8.4	320	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.7	28.4	26.1	27.8	25.4	27.2	24.8	27.2	28.6	26.5	28.0	25.8	27.2	4.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.8	22.4	28.3	26.0	21.4	27.7	25.4	20.5	27.1	85.7	28.7	82.5	28.0	79.8	27.4	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.1	13.5	11.2	DB	-3.7	28.3	2.5	0.8	-5.4	28.9	-6.9	29.4	-8.3	29.9	-10.1	30.5	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	7.7	7.4	9.0	11.9	15.8	21.5	25.1	25.2	22.9	18.7	13.9	10.7	(6)	
	DBStd	7.05	4.28	3.48	2.71	2.57	2.64	2.69	1.54	1.49	1.82	2.79	3.56	3.93	(7)	
	HDD10.0	292	97	83	51	11	0	0	0	0	0	0	9	41	(8)	
	HDD18.3	1617	331	306	290	192	87	7	0	0	0	29	137	237	(9)	
	CDD10.0	2434	24	11	19	68	179	346	468	472	388	271	127	62	(10)	
	CDD18.3	718	0	0	0	0	8	102	209	214	138	42	4	1	(11)	
	CDH23.3	3673	0	0	0	0	2	252	1436	1527	434	22	0	0	(12)	
(13)	CDH26.7	244	0	0	0	0	0	4	101	124	14	0	0	0	(13)	
(14) Wind	WSAvg	5.2	6.8	6.5	5.8	5.0	4.2	3.8	4.1	4.1	4.6	5.2	5.7	6.3	(14)	
(15) Precipitation	PrecAvg	1660	133	128	137	118	125	125	175	169	151	129	132	139	(15)	
	PrecMax	2049	250	262	217	202	244	229	255	317	291	224	203	230	(16)	
	PrecMin	1267	62	57	63	49	52	60	76	77	37	50	17	63	(17)	
	PrecStd	174	42	44	39	37	45	41	45	58	61	51	49	45	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.7	16.6	16.2	19.0	23.3	27.3	29.4	29.5	28.3	24.9	22.2	20.1	(19)	
		MCWB	16.7	15.4	14.5	17.7	21.9	25.8	27.2	27.1	26.1	23.5	20.5	18.8	(20)	
	2%	DB	16.4	14.5	14.9	17.7	21.8	26.1	28.4	28.6	26.8	24.0	21.0	18.4	(21)	
		MCWB	15.2	13.3	13.6	16.8	20.9	24.6	26.2	26.2	25.1	22.5	19.4	17.6	(22)	
	5%	DB	15.3	13.2	13.8	16.6	20.4	25.4	27.7	27.9	26.0	23.4	20.0	17.2	(23)	
		MCWB	14.0	11.5	12.6	15.8	19.7	23.8	25.5	25.3	24.2	21.6	18.3	16.2	(24)	
	10%	DB	13.9	12.2	12.8	15.6	19.3	24.8	27.2	27.3	25.4	22.6	18.8	16.2	(25)	
		MCWB	12.1	10.4	11.5	14.8	18.4	23.3	25.0	24.6	23.6	20.6	17.3	15.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.2	16.2	16.2	18.3	23.2	24.9	28.3	27.9	26.9	24.1	20.1	17.7	(27)	
		MCDB	17.5	16.4	16.3	18.7	23.9	25.3	29.4	29.3	28.1	24.6	20.6	18.0	(28)	
	2%	WB	16.2	14.4	14.9	17.0	21.9	24.3	27.2	27.4	26.3	23.0	18.7	16.8	(29)	
		MCDB	16.6	15.1	15.2	17.5	22.6	24.8	28.5	28.9	27.8	23.6	19.8	17.2	(30)	
	5%	WB	14.9	12.6	13.9	16.1	21.2	23.9	26.4	26.7	25.5	21.7	17.8	15.7	(31)	
		MCDB	15.6	13.5	14.4	16.6	21.9	24.6	27.7	28.2	26.7	22.8	18.9	16.2	(32)	
	10%	WB	13.4	11.0	12.5	15.1	20.3	23.4	25.7	26.0	24.7	20.7	16.8	14.7	(33)	
		MCDB	14.5	12.3	13.4	15.6	21.0	24.2	27.0	27.6	25.8	22.0	17.8	15.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	4.6	3.8	3.1	2.6	2.1	1.8	1.7	1.8	2.5	3.4	4.0	(35)	
		MCDBR	5.6	6.2	4.9	3.8	2.8	2.2	2.0	2.0	2.0	2.5	4.0	4.8	(36)	
	5% WB	MCWBR	6.1	7.6	5.1	3.9	2.8	2.4	2.6	2.6	2.5	2.4	5.5	6.2	(37)	
		MCWBR	5.3	7.0	5.0	3.7	2.8	1.7	2.0	1.9	2.0	2.2	4.6	4.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.334	0.342	0.358	0.400	0.445	0.479	0.503	0.495	0.417	0.373	0.341	0.337	(40)	
		taud	2.505	2.482	2.464	2.352	2.260	2.216	2.157	2.180	2.391	2.494	2.528	2.520	(41)	
	Edn at Noon		857	895	909	883	844	812	789	787	837	848	843	829	(42)	
			81	93	104	122	136	142	150	144	111	92	79	76	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.15	3.00	4.18	5.44	6.16	6.63	6.28	5.70	4.69	3.68	2.56	1.99	(44)	
	RadStd		0.17	0.29	0.34	0.34	0.46	0.39	0.29	0.40	0.33	0.30	0.21	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+166	+107

Nomenclature: See separate page